

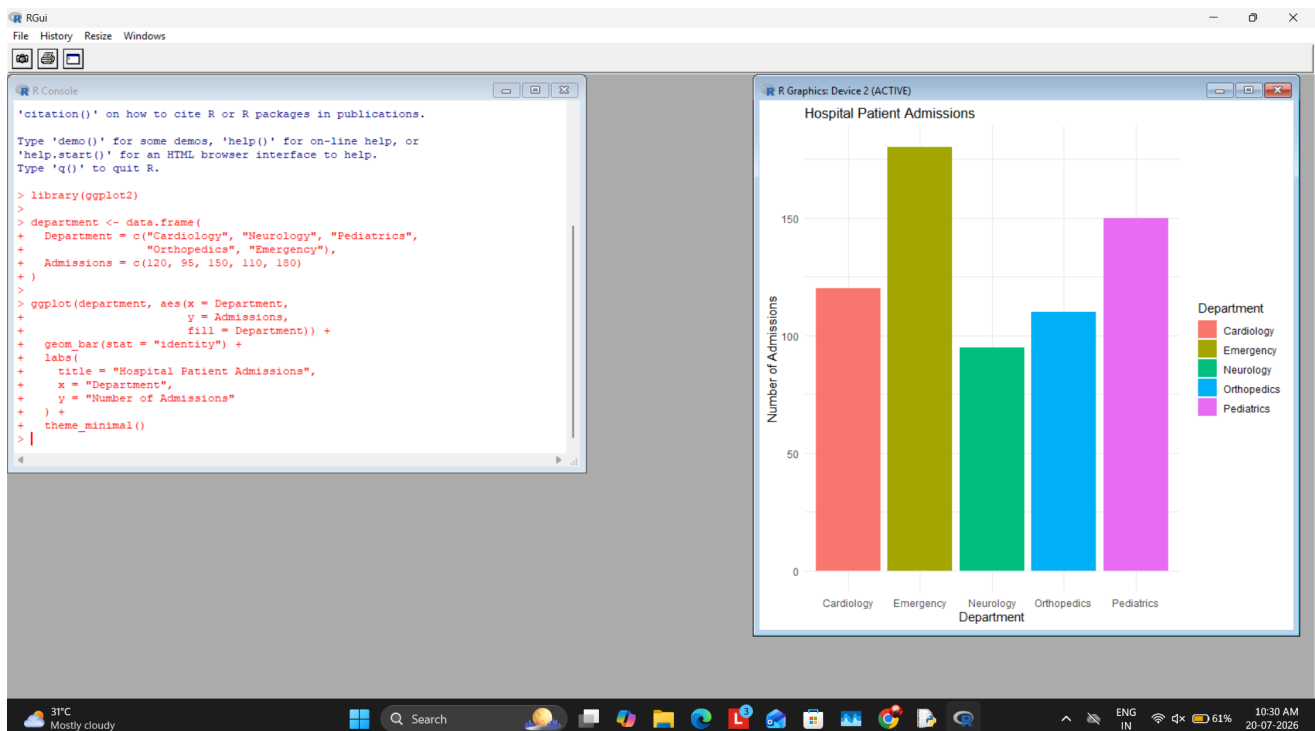
R Programs with Output Screenshots (Q1-Q5)

Q1 Hospital Patient Admission Analysis

```
library(ggplot2)

department <- data.frame(
  Department=c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),
  Admissions=c(120,95,150,110,180)
)

ggplot(department,aes(x=Department,y=Admissions,fill=Department))+
  geom_bar(stat="identity")+
  labs(title="Hospital Patient Admissions",
       x="Department",
       y="Number of Admissions")+
  theme_minimal()
```



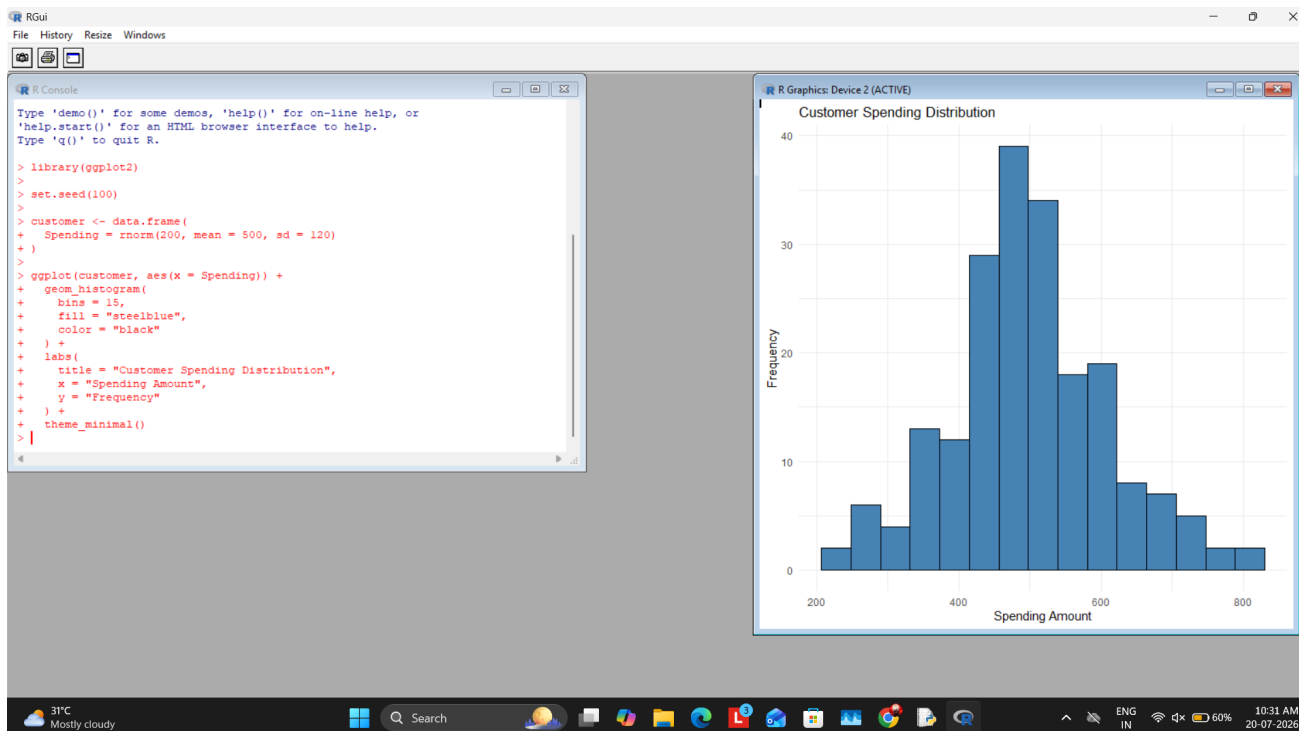
Q2 E-Commerce Customer Spending Behaviour

```
library(ggplot2)

set.seed(100)

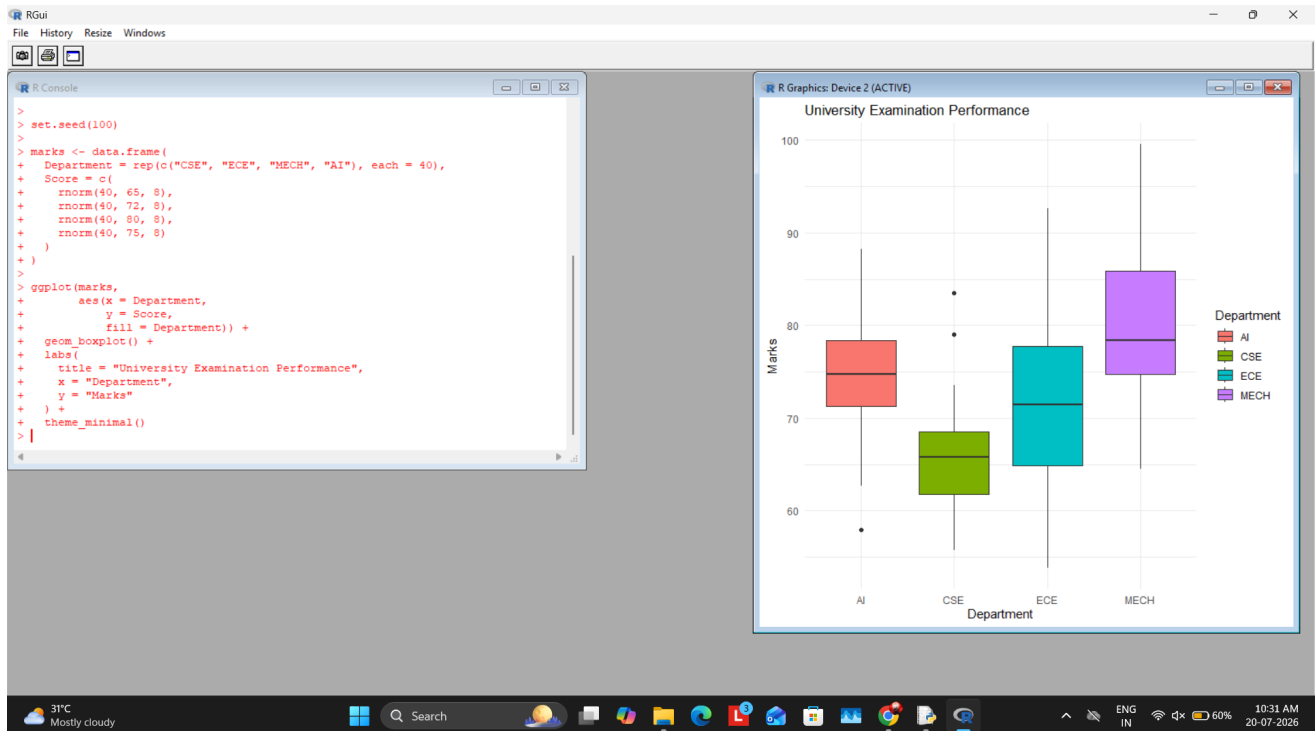
customer<-data.frame(
  Spending=rnorm(200,mean=500,sd=120)
)

ggplot(customer,aes(x=Spending))+
  geom_histogram(bins=15,fill="steelblue",color="black")+
  labs(title="Customer Spending Distribution",
       x="Spending Amount",
       y="Frequency")+
  theme_minimal()
```



Q3 University Examination Performance Dashboard

```
library(ggplot2)  
  
set.seed(100)  
  
marks<-data.frame(  
  Department=rep(c("CSE", "ECE", "MECH", "AI"), each=40),  
  Score=c(rnorm(40,65,8),  
          rnorm(40,72,8),  
          rnorm(40,80,8),  
          rnorm(40,75,8))  
)  
  
ggplot(marks,aes(x=Department,y=Score,fill=Department))+  
  geom_boxplot()+  
  labs(title="University Examination Performance",  
        x="Department",  
        y="Marks")+  
  theme_minimal()
```



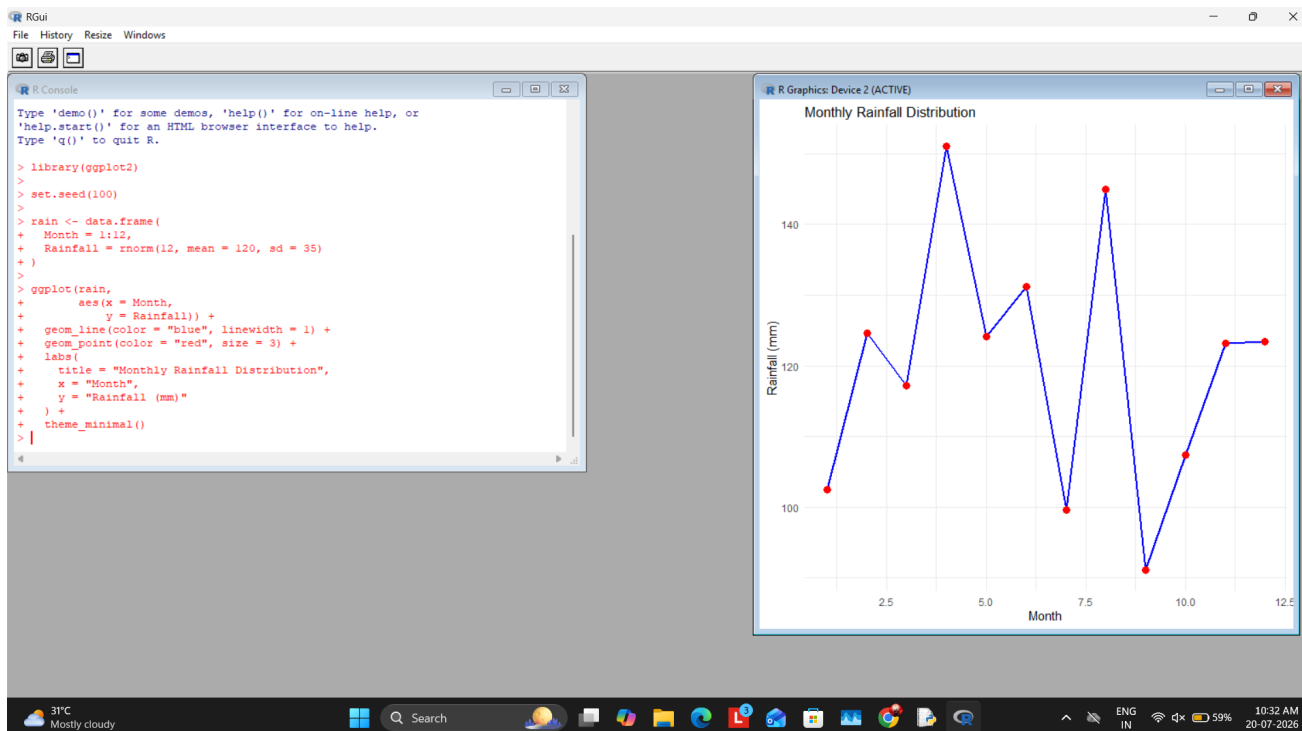
Q4 Climate and Rainfall Distribution Study

```
library(ggplot2)

set.seed(100)

rain<-data.frame(
  Month=1:12,
  Rainfall=rnorm(12,mean=120,sd=35)
)

ggplot(rain,aes(x=Month,y=Rainfall))+
  geom_line(color="blue",linewidth=1)+
  geom_point(color="red",size=3)+
  labs(title="Monthly Rainfall Distribution",
    x="Month",
    y="Rainfall (mm)")+
  theme_minimal()
```



Q5 Bank Loan Risk Assessment

```
library(ggplot2)
```

```
set.seed(100)
```

```
loan<-data.frame(  
  LoanAmount=rgamma(200,shape=2,scale=5000)  
)
```

```
ggplot(loan,aes(x=LoanAmount))+  
  geom_histogram(bins=15,fill="tomato",color="black")+  
  labs(title="Loan Amount Distribution",  
        x="Loan Amount",  
        y="Frequency")+  
  theme_minimal()
```

